Shubhankar Nehra

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Education

Virginia Tech — Blacksburg, VA
B.S. in Computer Science, Minor in Cybersecurity | GPA: 3.63 — *Aug 2024 – Present*

Experience

Faith Innovations - Full-Stack Developer — Remote
Faith CRM – Medical Equipment Business — *May 2026 – Present*

- Architected and built a full-stack CRM (customer relationship management) platform (**Next.js 15, TypeScript, Supabase/PostgreSQL**), replacing manual spreadsheet-based sales tracking.
- Implemented **Postgres Row-Level Security (RLS)** policies across three distinct roles to enforce data access control at the database layer.
- Designed two configurable sales pipelines and an external cron scheduler to automate recurring tasks and reminders.
- Planned and executed a data migration off the legacy Bigin Zoho CRM system, developing an **"import with unclassified states"** strategy to preserve historical records without forcing premature reclassification.

Spacenet - Undergraduate Research Assistant — Blacksburg, VA
Backend Developer — *Feb 2026 – May 2026*

- **Decentralized system architecture** by transitioning from user-based authentication to direct resource access model.
- Performed static code and package analysis to resolve cross-platform dependencies, **ensuring environment agnostic deployment** across Windows and Linux host systems.

DBWorkout - Undergraduate Research Assistant — Blacksburg, VA
Backend Developer - Secure Authorization — *Aug 2025 – Feb 2026*

- **Led a Defense-in-Depth Security hardening initiative** by creating global exception handling, mitigating information leaks, and preventing sensitive stack traces.
- **Engineered a secure error tracing mechanism** that intercepts fatal system crashes and generates a sanitized JSON response along with a unique UUID, enabling the dev team to internally log errors without exposing backend logic, and have a foundation for providing user support in the future.
- **Refactored** and extended the current **RBAC architecture** to be future proof, scalable, and maintainable for newer roles.

Projects & Achievements

Cyber Apocalypse 2026 CTF | *HackTheBox (Team)* — *July 2026*

- Placed **571/6744 (Top 9%)** globally, competing in web exploitation and reverse engineering.
- Exploited a broken session management flaw to hijack authenticated sessions in a web challenge (Gatery).
- Achieved stored XSS via a **Google APIs JSONP-based CSP bypass**, defeating the target's content security policy (Massagold).

CancerScan | *Python, TensorFlow, OpenCV (Team)* — *Jul 2025 – Aug 2025*

- Trained a **CNN (ResNet-50 transfer learning)** for cancer detection on a Brain MRI dataset, achieving **92% validation accuracy** through hyperparameter tuning and data augmentation.
- Garnered interest from a hospital in Greater Noida, India, for a custom deployment of the tool.

Certifications

CompTIA A+ — *Dec 2025*

ISC² Certified in Cybersecurity (CC) — *Aug 2025*

Technical Skills

Languages: C/C++, Python, Assembly (RISC V), SQL, Java, TypeScript
Security & Networking: Active Directory, RBAC, Wireshark, Ghidra
Developer Tools & Frameworks: GDB, Docker, Git, Virtual Box/VMWare, Rars, Ripes, Flask, Next.js, Supabase/PostgreSQL